VI SEMESTER

Natural Language Processing and Transformers - AI363IA

SEE – Project Based Learning

Phase-I Review

on

WILDAI: Wildlife Intelligence & Language-driven Decision Agent Interface

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Introduction

Wildlife conservation **policies** are **critical for protecting biodiversity**, endangered species, and ecological balance. However, **these policies are typically long, complex, and written in legal language,** making them **difficult to interpret** quickly.

With advancements in **Natural Language Processing and Generative AI**, it is possible to automatically **analyze** and **extract** meaningful insights from such documents.

This project proposes **an intelligent system that combines NLP** techniques with **a simple RAG AI framework** to analyze wildlife policies and generate structured insights such as summaries, key entities, and recommendations.

Existing System

Currently, wildlife policy analysis is mostly done manually by researchers, policymakers, and environmental organizations such as the World Wildlife Fund and the International Union for Conservation of Nature.

Existing digital tools provide:

- **Basic keyword** search
- **Manual reading** of PDFs

Limitations:

- Time-consuming
- No automation in extracting insights
- No intelligent interpretation or recommendations
- Lack of interactive querying

Names of existing systems : SMART(Spatial Monitoring and Reporting Tool) , EarthRanger , Marxan ,etc.

Identification of problem

The key issues in the current system are :

- Difficulty in understanding lengthy policy documents
- Lack of **automated extraction** of **important information**
- **No system to identify key entities (species, laws, regions)**
- **No intelligent system** to **generate summaries or insights**
- **Absence of AI-driven decision support.**

Literature Review

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Author and Title of paper	Summary	Gaps & Limitations
S. Gupta, A. Ghogale, A. Farooqui, S. Mhatre and S. Dubey, "Wildlife Analysis and Conservation System," 2025 International Conference on Emerging Trends in Industry 4.0 Technologies (ICETI4T), Navi Mumbai, India, 2025, pp. 1-6, doi: 10.1109/ICETI4T63625.2025.11132266.	The system uses machine learning to monitor wildlife, track animal movements, and detect threats like poaching in real time. It helps generate alerts and provides useful data for conservation efforts.	It depends on large datasets and costly infrastructure, has limited coverage, and may face delays or accuracy issues in real-world environments.
M. Cybi, E. Joy, S. Evangeline, J. Paul and S. Sanjai, "Wildlife Detection and Behavioral Analysis in Farmland using YOLOv8m and LSTM," 2025 3rd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS), Erode, India, 2025, pp. 246-251, doi: 10.1109/ICSSAS66150.2025.11080886.	The system uses deep learning (YOLOv8m, DeepSORT, and LSTM) to detect, track, and analyze wildlife behavior in farmland, achieving high accuracy and enabling real-time alerts. It helps distinguish between harmless movements and potential threats to crops, improving monitoring efficiency and response time.	It relies on a limited set of animal classes and a custom dataset, may struggle with unseen species or extreme conditions, and requires significant computational resources. Additionally, it focuses on detection and behavior analysis without integrating broader conservation policies or large-scale ecosystem insights.

Literature Review

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Research Paper Name	Overview	Gaps & Limitations
A. Walia, A. Ghosh, V. Kumar, A. Prakash and G. P, "Wildlife Conservation and Habitat Monitoring," 2024 4th International Conference on Innovative Sustainable Computational Technologies (CISCT), Dehradun, India, 2024, pp. 1-5, doi: 10.1109/CISCT62494.2024.11134181.	The study supports SDG 15 by using big data tools like Apache Spark and PySpark MLlib to analyze species distribution and habitat suitability from the GBIF dataset. It applies Decision Tree classification and K-means clustering to identify optimal conservation areas, with Tableau visualizations aiding real-time decision-making.	The approach depends heavily on dataset quality and geographic features, lacks integration of textual policy insights, and may not fully capture dynamic environmental changes or real-time field conditions.
M. Hossain et al., "Data Analytics in Wildlife Management Business Insights and Technological Solutions," 2024 International Conference on Progressive Innovations in Intelligent Systems and Data Science (ICPIDS), Pattaya, Thailand, 2024, pp. 539-544, doi: 10.1109/ICPIDS65698.2024.00089.	The study highlights how data analytics, machine learning, and predictive modeling can improve wildlife management by identifying patterns in population dynamics, migration, and threats. It enables better decision-making, resource allocation, real-time monitoring, and proactive conservation strategies, including anti-poaching efforts.	The approach depends heavily on large-scale, high-quality data and advanced infrastructure, which may not be available in all regions. It also lacks integration with policy analysis and may face challenges in real-time deployment, model interpretability, and adapting to rapidly changing environmental conditions.

Summary of Literature Review

NLP techniques have been successfully applied in domains like legal and policy analysis, as well as wildlife monitoring using machine learning and big data analytics.

Existing systems mainly focus on tracking, habitat analysis, and predictive modeling, providing valuable insights for conservation and decision-making.

However, the literature also highlights key gaps, including the **lack of integration between textual policy analysis and data-driven wildlife monitoring systems**.

VVI : Most approaches do not combine NLP with real-time insights or intelligent decision-making. This creates an opportunity to develop an integrated system that uses NLP, generative AI, and agentic approaches to analyze wildlife conservation policies effectively.

Problem Definition

Develop an automated system using Natural Language Processing and Generative AI that can process policy documents, extract meaningful insights, generate concise summaries, and **support decision-making through an intelligent, rag-based approach.**

Objectives

- To preprocess and clean policy documents
- To perform Named Entity Recognition (NER) for extracting:
  - Species
  - Locations
  - Laws and organizations
- To classify policies based on conservation focus
- To generate summaries using **transformer models**
- **To implement a simple RAG AI system that:**
  - **Decides which NLP tasks to apply**
  - **Produces a final structured report**

Methodology

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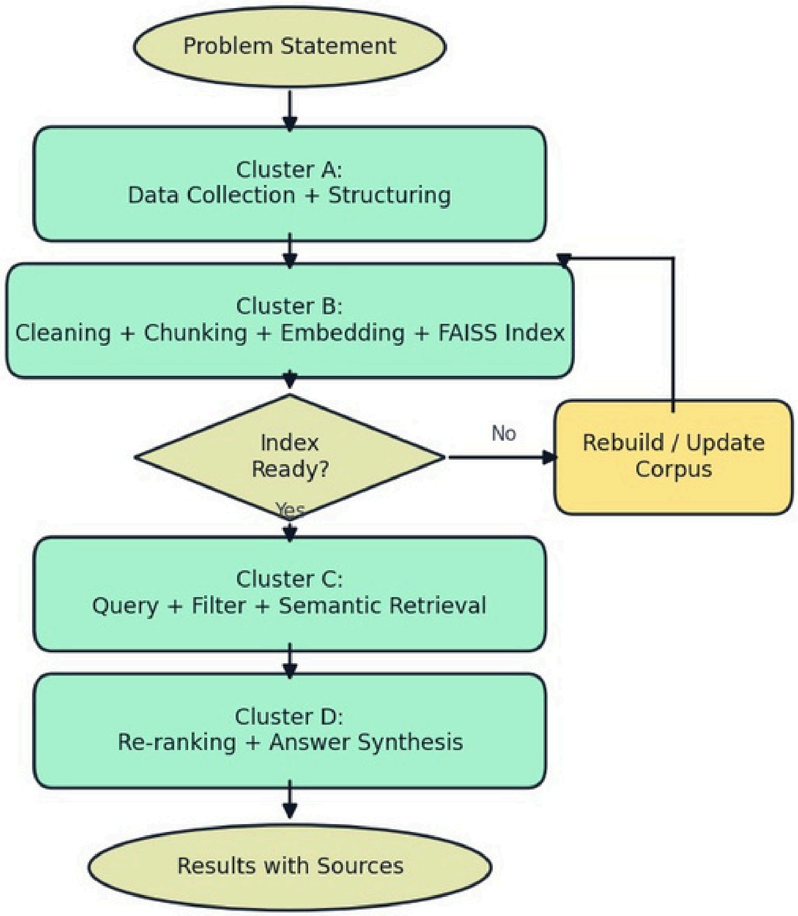
1.Data Collection and Structuring

- Collected high-authority wildlife data from policies, legal acts, treaties, species, and ecosystem sources.
- Standardized records with metadata: category, source, year, tags, and URL.
- Built a large curated corpus: 364 documents, 34 categories, 21 sources (1960-2026).

2.Preprocessing and Semantic Indexing

- Cleaned and normalized text, then split into semantic chunks (sentence-aware chunking).
- Converted chunks into embeddings using all-mpnet-base-v2 (768-d).
- Indexed embeddings in FAISS(Facebook AI Similarity Search) for fast retrieval (206,529 chunks, ~605 MB index).



Methodology

3.Query Understanding and Retrieval

- User submits query with optional filters (year, category, source).
- System performs semantic search in FAISS and fetches top relevant chunks.
- Re-ranking combines semantic score + keyword overlap + temporal weighting.

4.Answer Generation and Validation

- Deduplicates and prioritizes evidence (especially recent policy docs for “latest” queries).
- Synthesizes a final response with key insights, year context, and source attribution.
- Delivers output through FastAPI + React interface with highlighting and filters.
- Result: A fast, explainable RAG workflow for wildlife policy and conservation research

Proposed tools and techniques

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Programming Language

- Python

NLP Techniques

- Tokenization
- Stopword removal
- Lemmatization
- Named Entity Recognition (NER)

Machine Learning / Models

- TF-IDF (Term Frequency-Inverse Document Frequency)+ Logistic Regression (for classification)
- Transformer models

T5(Text-to-Text Transfer Transformer)

BART(Bidirectional and Auto-Regressive Transformers)

Libraries

- SpaCy
- NLTK
- Hugging Face Transformers

Agentic AI Approach

- Rule-based or LLM-based decision system
- Selects tasks like:
    - Summarization
    - Entity extraction
    - Classification

Interface

- React

Expected Outcomes

The system will:

- Accept wildlife policy documents as input (from every years)
- Generate:
  1. Concise summary
  2. Extracted key entities
  3. Policy category (organizes policies, instruments, or rules based on shared characteristics, intent, or subject matter (e.g., HR, Security, Data Privacy.))
  4. AI-generated recommendations

Example output:

- Summary of policy
- List of protected species
- Identified conservation strategy
- Suggested improvements

This will significantly reduce manual effort and improve policy understanding.

Conclusion

The proposed Wildlife Conservation Policy Analyzer demonstrates how modern AI techniques such as Natural Language Processing, Generative AI, and Agentic AI can be applied to real-world environmental challenges.

By automating the analysis of policy documents, the system provides faster, more accurate, and structured insights, which can assist policymakers, researchers, and conservation organizations in making informed decisions.

This project also highlights the practical integration of AI into governance and sustainability domains, making it a relevant and impactful application for academic and real-world use.

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References

[1] J. Devlin et al., “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” NAACL, 2019.

[2] C. Raffel et al., “Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer,” JMLR, 2020.

[3] M. Lewis et al., “BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation,” ACL, 2020.

[4] R. Mihalcea and P. Tarau, “TextRank: Bringing Order into Text,” EMNLP, 2004.

[5] M. Honnibal et al., “spaCy: Industrial-strength Natural Language Processing in Python,” 2020.

[6] G. Salton and C. Buckley, “Term-Weighting Approaches in Automatic Text Retrieval,” 1988.

[7] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” NeurIPS, 2020.

[8] I. Chalkidis et al., “Legal-BERT: The Muppets Straight Out of Law School,” Findings of EMNLP, 2020.

[9] T. Brown et al., “Language Models are Few-Shot Learners,” NeurIPS, 2020.

[10] H. Chase, “LangChain: Building Applications with Large Language Models,” 2023.

[11] A. Vaswani et al., “Attention is All You Need,” NeurIPS, 2017.

[12] K. Clark et al., “ELECTRA: Pre-training Text Encoders as Discriminators Rather Than Generators,” ICLR, 2020.

[13] Y. Liu et al., “RoBERTa: A Robustly Optimized BERT Pretraining Approach,” 2019.

[14] J. Peters et al., “Deep Contextualized Word Representations,” NAACL, 2018.

[15] T. Mikolov et al., “Efficient Estimation of Word Representations in Vector Space,” 2013.

References

[16] J. Pennington et al., “GloVe: Global Vectors for Word Representation,” EMNLP, 2014.

[17] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” 1997.

[18] Y. Goldberg, “Neural Network Methods for Natural Language Processing,” 2017.

[19] S. Bird et al., “NLTK: The Natural Language Toolkit,” 2009.

[20] Q. Le and T. Mikolov, “Distributed Representations of Sentences and Documents,” ICML, 2014.

[21] A. Radford et al., “Improving Language Understanding by Generative Pre-Training,” 2018.

[22] S. Ruder, “Neural Transfer Learning for Natural Language Processing,” 2019.

[23] Y. Zhang et al., “Deep Learning Based Text Classification: A Comprehensive Review,” 2018.

[24] Data Analytics in Wildlife Management Business Insights and Technological Solutions,2024

[25] Wildlife Conservation and Habitat Monitoring , 2024

[26] Wildlife Analysis and Conservation System ,2025.

[27] Wildlife Detection and Behavioral Analysis in Farmland using YOLOv8m and LSTM , 2025

Thank You